Higher National Diploma in Engineering (Electrical and Electronics)

First Year, First Semester Examination – 2019

EE1107 - ELECTRICAL PRINCIPLES – A

Answer Five question only Time : 2 hours

Question 1

i. Determine the current through the 4Ω resistor in the circuit given in Figure 1:

a) by applying the Kirchhoff's law. (5 marks)

b) by applying Thevenin's theorem. (8 marks)

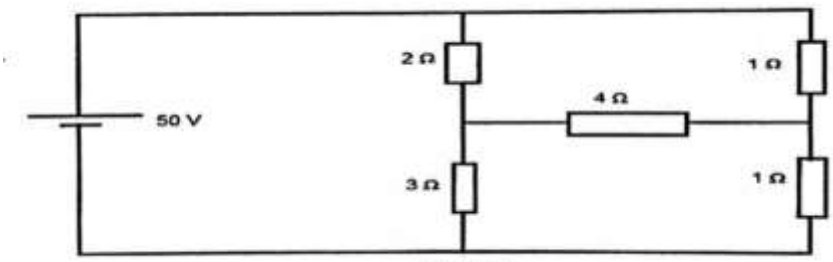

Figure 1

ii. By applying the **Superposition theorem** for the circuit given in Figure 2, prove that the current through $R_2\Omega$ resistor is: (7 marks)

$$\left[\frac{R_3 \cdot E_1}{R_1 \cdot (R_2 + R_3) + R_2 \cdot R_3}\right] + \left[\frac{E_2 \cdot (R_1 + R_3)}{R_2 \cdot (R_1 + R_3) + R_1 \cdot R_3}\right]$$

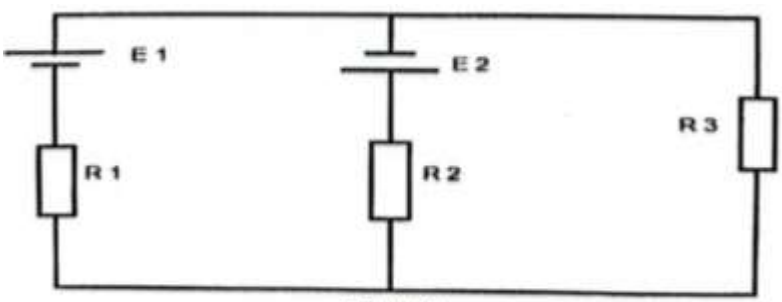

Figure 2

Question 2

i. Mention why **watt-second** is not used as a unit to measure the electrical energy practically. (2 marks)

ii. A 100 watts power is developed in a heating coil when a constant voltage is applied at the two ends of it. This heating coil is cut into two equal sections and they are joined in parallel and connected across the same voltage. Find the **power developed in new arrangement**. (5 marks)

iii. Why is the conductor **heated** when electrical current passes through it? (2 marks)

iv. State the **3 effects of electrical current**. (3 marks)

v. State the **3 factors** which determine the amount of heat required to heat a material. (3 marks)

vi. An electric kettle marked $1\mathrm{kW}, 230$ volts takes 5 minutes to bring $1\mathrm{kg}$ of water at $15^{\circ}\mathrm{C}$ to boiling point $100^{\circ}\mathrm{C}$. Find the **efficiency of the kettle**. The specific heat capacity of water is $4187\ \mathrm{Jkg}^{-1}\mathrm{k}^{-1}$. (5 marks)

Question 3

i. Name **3 practical capacitors**. (3 marks)

ii. What is the **voltage across the capacitor plates** when it is fully discharged? (1 mark)

iii. State the factors which determine the **capacitance of capacitor** is dependent upon. (3 marks)

iv. Mark the following on the **charging current curve of the capacitor**: a. Maximum current through the capacitor, b. Time constant, c. The time period in which the capacitor is almost fully charged. (4 marks)

v. Find the **voltage across the $5\mu\mathrm{F}$ capacitor** of the circuit given in Figure 3. (4 marks)

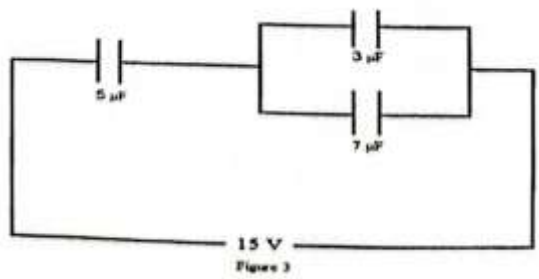

Figure 3

vi. A $10\mu\mathrm{F}$ capacitor initially charged to a potential difference of $400\ \mathrm{volts}$ is discharged through an unknown resistance. After one minute the potential difference at the terminals of the capacitor is $200\ \mathrm{volts}$. Calculate the **value of resistance** through which the capacitor is discharged. (5 marks)

Question 4

i. Define the term **magnetic flux** in a magnetic field. (1 mark)

ii. What is **mutually induced electromotive force (emf)**, explain briefly with the help of a diagram [it is required to indicate the current direction of unexcited coil clearly]. (3 marks)

iii. Indicate the **residual flux density** and **coercive force** on the hysteresis loop. (3 marks)

iv. An iron ring $5\mathrm{cm}$ in radius and $5\mathrm{cm}^2$ in cross section is wound with 300 turns. An exciting current of $1.2\mathrm{A}$ is required to produce a flux density of $1.2\ \mathrm{T}$. Calculate the **energy stored in the magnetic field**. (3 marks)

v. Show the **time constant** on the inductive circuit discharging current curve. (2 marks)

vi. A coil having self-inductance of $2\mathrm{H}$ and resistance 4Ω is connected to a constant $200\mathrm{V}$ supply source. How long does it take the **voltage across the resistance to reach** $75\mathrm{V}$? (3 marks)

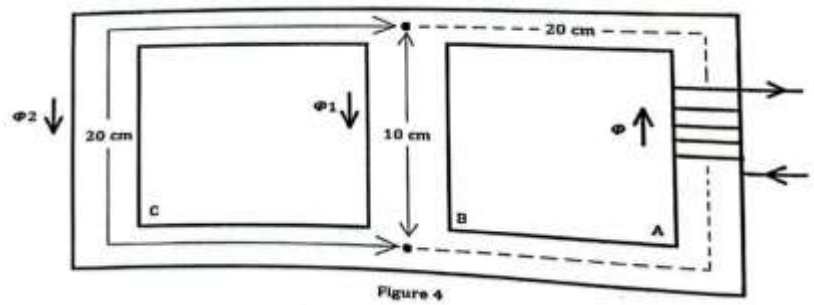

Figure 4

vii. A cast steel magnetic structure made up of a bar of cross sectional area of $2\mathrm{cm}^2$ is given in the figure (4). Determine the **current that the 100 turns magnetizing coil** on the right limb should carry. A flux of $2\mathrm{mWb}$ is produced in the extreme left limb. Relative permeability of the cast steel is 600 and neglect the leakage. (5 marks)

Question 5

i. Draw the **corresponding waveforms** to the given phasor diagrams in the figure (5). (2 marks)

Figure 5

ii. Find the **rms value** for the given maximum voltage of $282.84\mathrm{V}$. (1 mark)

iii. Draw the **power triangle** and indicate the 3 powers on it. (3 marks)

iv. Inductive reactance and impedance for a.c. R-L series circuit are 14Ω and 20Ω respectively. Inductance of the coil is $1\mathrm{H}$. Determine: a. **Resistance** of the circuit, b. **Power factor** of the circuit [it must be indicate whether the power factor is leading or lagging], c. **Frequency** of applied voltage. (4 marks)

v. For the network shown in figure (6) determine **active power developed in the capacitor** and **total active power developed in the circuit**. (6 marks)

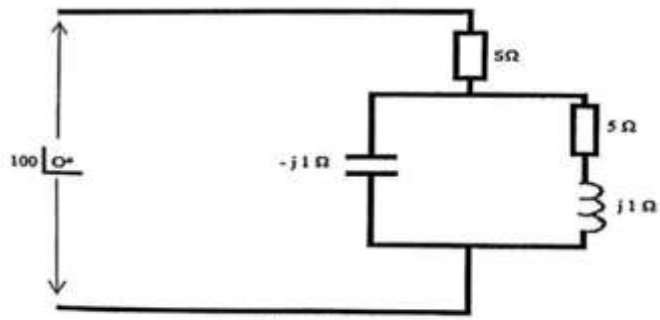

Figure 6

vi. Write down the expression for **resonant frequency** of series R-L-C resonant circuit. (1 mark)

vii. A coil having a resistance of 25Ω and inductance of 100mH is connected in series with a capacitance of $10\mu\text{F}$ across a 200V variable frequency supply. Find: a. the **resonant frequency**, b. the **current at resonance**. (3 marks)

Question 6

i. State **two possible phase sequences** if phases are R, Y and B. (1 mark)

ii. Current in each line is, the vector difference of each phase currents in balanced delta connection. Draw the **complete phasor diagram** on this connection. [Hint: Line voltages lead respective phase currents by an angle θ]. (4 marks)

iii. Find the **phase angle between line voltage and line current** with the help of phasor diagram drawn in part ii of this question. (2 marks)

iv. Write down **four (04) differences between star and delta connections**. (4 marks)

v. A star connected three phase load consisting of 3 similar impedances. When the load is connected to a three phase $400\text{V}, 50\ \text{Hz}$ supply. The line current is $14.43\ \text{A}$ and the power factor is $0.8\ \text{lagging}$. a. Calculate the **total power taken by the load**. (1 mark) b. Calculate the **resistance and inductance of each phase** of the load. (3 marks) c. Draw the **relevant phasor diagram**. (4 marks) d. Find the **angle between line voltage and line current**. (1 mark)